Sequences and Functions 2

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Contents

1 Functions **1**
1.1 Power Series 1
1.2 Exponential and Trigonometic Functions 3
1.3 Limits of Functions 4

2 Continuity **7**
2.1 Composition of Continuous Functions 8
2.2 Continuity of Selected Functions 8
2.3 The Intermediatve Value Theorem 9
2.4 The Weierstrass Extreme Value Theorem 9
2.5 The Natural Logarithm 9

3 Derivatives **11**
3.1 The Algebra of Derivatives 12
3.2 The Chain Rule 13
3.3 The derivative of inverse functions 14

1 Functions

1.1 Power Series

Definition. A **power series** is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n.$$

where $a_0, a_1, ...$ are all real numbers, called the coefficients of the power series.

Theorem. For any power series $\sum_{n=0}^{\infty} a_n x^n$ either

1. $\sum_{n=0}^{\infty} |a_n x^n|$ converges $\forall x \in \mathbb{R}$
2. $\sum_{n=0}^{\infty} |a_n x^n|$ diverges $\forall x \in \mathbb{R} \setminus \{0\}$
3. $\exists R \in (0, \infty)$ such that $\sum_{n=0}^{\infty} |a_n x^n|$ converges if $|x| < R$ and diverges

if $|x| > R$.

Proof. Suppose their exists at least one $x > 0$ such that the power series converges and at least one $z > 0$ such that it doesn't.

Define $S := \{r \geq 0 \mid (|a_n|r^n)_{n\in\mathbb{N}_0} \text{ is bounded}\}$ and $R := \sup S$. Hence $x \in S$. Suppose there exists $r > z \in S$. Then

$$|a_n z^n| = \frac{|a_n|z^n}{|a_n|r^n}|a_n|r^n = (|a_n|r^n)\left(\frac{z^n}{r^n}\right) \leq M\left(\frac{z}{r}\right)^n$$

for some M. Since this is less than a geometric sequence, the sum must converge, a contradiction. This means z is an upper bound of S.

For any $x \in S$, we can choose a $x < r < R$. Then by the same argument as above, $\sum_{n=1}^{\infty} |a_n x^n|$ is less than a geometric series, so converges. If $x > R$, then $a_n x^n$ is unbounded and clearly does not converge.

□

Remark. In Case 3, the sequence may or may not converge if $|x| = R$, and it depends on the sequence $(a_n)_{n\in\mathbb{N}}$.

Definition. (Radius of Convergence)

- If $\sum_{n=0}^{\infty} |a_n x^n|$ converges $\forall x \in \mathbb{R}$, we define the **radius of convergence** to be ∞
- If $\sum_{n=0}^{\infty} |a_n x^n|$ diverges for all $|x| > 0$ we define the radius of convergence to be 0.
- If $\sum_{n=0}^{\infty} |a_n x^n|$ converges $|x| < R$ and diverges $|x| > R$, then the **radius of convergence** is defined as R.

Although we know the radius of convergence exists, we know nothing about it. The next theorem gives us some information about it.

Theorem. For any power series $\sum_{n=0}^{\infty} |a_n x^n|$, if

- $a_n \neq 0 \quad \forall n \in \mathbb{N}_0$ and $R = \lim_{n\to\infty} \frac{|a_n|}{|a_{n+1}|}$ exists, then R is the radius of convergence.

Proof. By the ratio test

$$\frac{a_{n+1}x^{n+1}}{a_n x^n} = x\left(\frac{a_{n+1}}{a_n}\right)$$

If $R = \infty$, this always converges to 0, so the power series is convergent for all x. Conversely, if $R = 0$, the ratio test gives $\infty > 1$ for all x. If $R = R$,

then

$$\begin{aligned}\lim_{n\to\infty} x\left(\frac{a_{n+1}}{a_n}\right) &< 1\\ \frac{x}{R} &< 1\\ x &< R.\end{aligned}$$

□

1.2 Exponential and Trigonometic Functions

A power series with radius of convergence $R \in (0, \infty]$ gives a function $f : (-R, R) \to \mathbb{R}$. Many important functions can be defined in this way.

Definition. For $x \in \mathbb{R}$, define

$$\exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Proposition. The power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ has infinite radius of convergence, thus the exponential function has domain equal to $\mathbb{R}$.

Proposition.

$$\exp(x+y) = \exp(x)\exp(y).$$

Proof. Non-examinable. □

Corollary. For any $x \in \mathbb{R}$,

- $\exp(x) > 0$
- $\exp(-x) = \frac{1}{\exp(x)}$.

Proof.

- We know $\exp(0) = 1$. Then

$$\begin{aligned}\exp(x-x) &= 1\\ \exp(x)\exp(-x) &= 1\exp(-x) &&= \frac{1}{\exp(x)}.\end{aligned}$$

- If $x \geq 0$, $\exp(x) \geq 1 > 0$. Then

$$\exp(-x) = \frac{1}{\exp(x)} > 0.$$

□

Definition. For all $x \in \mathbb{R}$,

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

Remark. We cannot find the radius of convergence using the ratio test, as many terms are equal to 0. However, we can compare these series with the exponential function and show they are bounded by the comparison test.

1.3 Limits of Functions

Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$.

Definition. Let $S \subseteq \mathbb{R}$ and $c \in \mathbb{R}$. We say c is a **cluster point** of S if there exists $(x_n)_{n\in\mathbb{N}}$ in $S \setminus \{c\}$ such that $\lim_{n\to\infty} x_n = c$.

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Let $L \in \mathbb{R}$ and let c be a cluster point of D.

We say f **converges to** L as x tends to c, or $\lim_{x\to c} f(x) = L$ if

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x \in D \quad \text{s.t.}$$

$$\text{if } 0 < |x - c| < \delta \text{ then } |f(x) - L| < \epsilon$$

In the above definition, the definition of $f(c)$ is irrelevent.

Theorem. Let $c \in \mathbb{R}$ be a cluster point of $D \subseteq \mathbb{R}$. Let $f : D \to \mathbb{R}$ and $L \in \mathbb{R}$. Then the following are equivalent:

- $\lim_{x\to c} f(x) = L$
- For any sequence $(x_n)_{n\in\mathbb{N}}$ in $D \setminus \{c\}$

$$\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) = L.$$

Proof.

- Suppose $\lim_{x\to c} f(x) = L$ and $\lim_{n\to\infty} x_n = c$. Now fix $\epsilon > 0$. Then we obtain a δ where

$$0 < |x - c| < \delta \Rightarrow |f(x) - L| < \epsilon.$$

Using this δ, we obtain an N such that

$$\forall n > N \quad |x_n - c| < \delta.$$

Then since $x_n \neq c$ for all x, the conditions for $f(x_n) \to L$ are met.

- Since assuming an implication is difficult to work with, we instead prove the contrapositive:

$$\lim_{x \to c} f(x) \neq L \wedge x_n \to c \Rightarrow \forall \epsilon > 0,\ \exists N,\ \forall n \geq N \quad |f(x_n) - L| > \epsilon.$$

The negation of the limit of f gives us an ϵ where for any δ we can find an x such that $0 < |x - c| < \delta$ and $|f(x) - L| > \epsilon$. Then, using this ϵ, fix $\delta > 0$. Choose N such that

$$|x - c| < \delta$$

and then the conclusion follows.

□

Proposition. Let c be a cluster point of $D \subseteq \mathbb{R}$. Suppose there exists a sequence $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ in $D \setminus \{c\}$ such that

- $\lim_{n \to \infty} x_n = c$ and $\lim_{y \to \infty} = c$
- $\lim_{n \to \infty} f(x_n) = A$ and $\lim_{n \to \infty} f(y_n) = B$ with $A \neq B$.

Then we conclude $\lim_{x \to c} f(x)$ cannot exist.

Proof. Suppose the limit exists. Then since x_n and y_n have the same limit

$$\lim_{n \to \infty} f(x_n) = A = B = \lim_{n \to \infty} f(y_n)$$

which is a contradiction. □

Proposition. Let c be a cluster point of $D \subseteq \mathbb{R}$ and let $f : D \to \mathbb{R}$. If $\lim_{x \to c} f(x) = L_1$ and $\lim_{x \to c} f(x) = L_2$ then

$$L_1 = L_2.$$

Proof. $\lim_{x \to c} f(x) = L$ is equivalent to $f(x_n) \to L$ for $x_n \to c$. Then

$$\lim_{n \to \infty} f(x_n) = L_1 = L_2.$$

So $L_1 = L_2$ by uniqueness of limits. □

Theorem. (Algebra of limits for functions)

Suppose $D \subseteq \mathbb{R}$ and c is a cluster point of D. Let $f, g : D \to \mathbb{R}$ be two functions.

Suppose $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$.

Let $x \in \mathbb{R}$. Then

- $\lim_{x \to c} (f(x) + g(x)) = L + M$
- $\lim_{x \to c} \alpha f(x) = \alpha L$
- $\lim_{x \to c} f(x)g(x) = LM$
- If $M \neq 0$, then $\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{L}{M}$

Proof. By turning the limit of the function into a sequence limit, they can be easily proved using the algebra of limits. $\square$

Theorem. Suppose $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ is a cluster point of D, $f : D \to \mathbb{R}$ and $\lim_{x \to c} f(x) = L$. Then

- If $f(x) \leq M \quad \forall x \in D$, then $L \leq M$
- If $f(x) \geq M \quad \forall x \in D$, then $L \geq M$.

Proof. Consider WLOG $f(x) \leq M$. Then for $x_n \to \infty$

$$\lim_{n \to \infty} f(x_n) \leq M$$

so since limits preserve non-strict inequalities, $L \leq M$. $\square$

Theorem. Suppose $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ is a cluster point of D and $f, g, h : D \to \mathbb{R}$ with

$$f(x) \leq g(x) \leq h(x) \quad \forall x \in D.$$

and

$$\lim_{x \to c} f(x) = L = \lim_{x \to c} h(x).$$

Then $\lim_{x \to c} g(x) = L$.

Proof. Consider $x_n \to c$. Then

$$f(x_n) \leq g(x_n) \leq h(x_n)$$
$$\lim_{n \to \infty} f(x_n) \leq \lim_{n \to \infty} g(x_n) \leq \lim_{n \to \infty} h(x_n).$$

so by sandwhich theorem the limit of g at c is L. $\square$

Now we will consider the limit of a function to ∞.

Definition. Let $D \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $L \in \mathbb{R}$. Suppose there exists a sequence $(a_n)_{n\in\mathbb{N}}$ in D such that $\lim_{n\to\infty} a_n = \infty$.

We say f converges to L as $x \to \infty$ if

$$\forall \epsilon > 0 \quad \exists N \in \mathbb{R} \text{ s.t. } \forall x \in D, \quad x > N$$

$$|f(x) - L| < \epsilon.$$

Hence we write $\lim_{x\to\infty} f(x) = L$.

Definition. Let $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ be a cluster point of D and $f : D \to \mathbb{R}$

- We say f diverges to ∞ as $x \to c$ if $\forall M \in \mathbb{R}\ \exists \delta > 0$ such that $\forall x \in D$ with $0 < |x - c| < \delta$ we have $f(x) > M$. In this case, we write $\lim_{x\to c} f(x) = \infty$.
- We say f diverges to $-\infty$ as $x \to c$ if $\forall M \in \mathbb{R}\ \exists \delta > 0$ such that $\forall x \in D$ with $0 < |x - c| < \delta$ we have $f(x) < M$. In this case, we write $\lim_{x\to c} f(x) = -\infty$.

2 Continuity

Definition. Let $D \subseteq \mathbb{R}$, $c \in D$ and $f : D \to \mathbb{R}$. We say f is continuous at c if

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x \in D, \quad |x - c| < \delta$$

$$|f(x) - f(c)| < \epsilon.$$

We say f is **continuous** on D if this holds true at every $c \in D$.

Theorem. Let $D \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and c be a cluster point of D.

Then f is continuous at c if and only if

$$\lim_{x\to c} f(x) = f(c).$$

Proof. Suppose f is continuous at c. Fix $\epsilon > 0$ and obtain δ from continuity of f. Then for all x with $0 < |x - c| < \delta$, we have

$$|f(x) - f(c)| < \epsilon.$$

Thus f converges to $f(c)$ at ϵ. Now suppose $\lim_{x\to c} f(x) = f(c)$. Fix $\epsilon > 0$ and obtain δ from the limit of f. Then for all $0 < |x - c| < \delta$,

$$|f(x) - f(c)| < \epsilon.$$

For the case where $|x - c| = 0$, $x = c$ and $|f(c) - f(c)| = 0 < \delta$, so f is continuous. $\square$

Remark. The relation between continuity at c and $\lim_{x \to c} f(x) = f(c)$ does not hold if c is not a cluster point. However, functions are always continuous at points that are not cluster points of the domain.

Theorem. (Algebra of continuous functions)
Let $D \subseteq \mathbb{R}$, $c \in D$ and $\alpha \in \mathbb{R}$. Let $f, g : D \to \mathbb{R}$ be continuous at c, then

- $f + g$
- αf
- fg

are all continuous at c.
If $g(c) \neq 0$, then $\frac{f}{g}$ is also continuous at c.

2.1 Composition of Continuous Functions

Given two sets $D, E \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $g : E \to D$, we can define the composite function

$$(f \circ g)(x) := f(g(x)).$$

Theorem. (Continuity of Continuous Functions)
Let $D, E \subseteq \mathbb{R}$, and $c \in E$. Let $f : D \to \mathbb{R}$ and $g : E \to D$.
If g is continuous at c and f is continuous at $g(c)$, then $f \circ g$ is continuous at c.

2.2 Continuity of Selected Functions

Proposition. The following functions are all continuous.

- $f(x) = \alpha$ for $\alpha \in \mathbb{R}$.
- $f(x) = x$
- $f(x) = \exp(x)$
- $f(x) = \sinh(x)$
- $f(x) = \cosh(x)$
- $f(x) = \sin(x)$
- $f(x) = \cos(x)$
- $f(x) = \sec(x), g(x) = \tan(x)$ where $f, g : \mathbb{R} \setminus \{\frac{\pi}{2} + n\pi | n \in \mathbb{Z}\} \to \mathbb{R}$
- $f(x) = \csc(x), g(x) = \cot(x)$ where $f, g : \mathbb{R} \setminus \{n\pi | n \in \mathbb{Z}\} \to \mathbb{R}$

2.3 The Intermediatve Value Theorem

Let $D \subseteq \mathbb{R}$, $\alpha \in \mathbb{R}$ and $f : D \to \mathbb{R}$. Solving the equation $f(x) = \alpha$ can rarely be done explicitly. However, if f is continuous on $D = [a, b]$, and if α is between $f(a)$ and $f(b)$, it is plausible that $f(x) = \alpha$ has a solution in D.

Theorem. Let $a, b \in \mathbb{R}$ with $a < b$. Suppose

- $f : [a, b] \to \mathbb{R}$ is continuous
- $f(a) \leq \alpha \leq f(b)$ or $f(b) \leq \alpha \leq f(a)$.

Then there exists $c \in [a, b]$ such that

$$f(c) = \alpha.$$

2.4 The Weierstrass Extreme Value Theorem

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$.

- We say $c \in D$ is a **minimum point** of f in D if

$$f(c) \leq f(x) \quad \forall x \in D.$$

- We say $c \in D$ is a **maximum point** of f in D if

$$f(c) \geq f(x) \quad \forall x \in D.$$

We say f **attains** its maximum or minimum in D if a maximum or minimum point exists in D.

Theorem. Let $a, b \in \mathbb{R}$ with $a < b$. Suppose $f : [a, b] \to \mathbb{R}$ is continuous. Then

- there exists finite $L > 0$ such that

$$|f(x)| \leq L \quad \forall x \in [a, b].$$

- f attains its maximum and minimum in $[a, b]$.

2.5 The Natural Logarithm

Before we can define the inverse of the exponential function, we must show that it is a bijection.

Proposition. The exponential function is bijective.

Proof. We first prove injectivity and then surjectivity.

- Let $x, y \in \mathbb{R}$ with $x > y$. Then

$$\begin{aligned}\frac{\exp(x)}{\exp(y)} &= \exp(x-y) \\ &= 1 + \sum_{n=1}^{\infty} \frac{(x-y)^n}{n!} > 1.\end{aligned}$$

  Therefore, as the sum is positive, $\exp(x) > \exp(y)$. This implies that $\exp : \mathbb{R} \to (0, \infty)$ is injective.

- Fix $y > 0$. Then

$$\exp(y) = 1 + y + \sum_{n=2}^{\infty} \frac{y^n}{n!} > y.$$

  This implies, for $y > 0$, $\exp(\frac{1}{y}) > \frac{1}{y} \Rightarrow \frac{1}{\exp(y)} = \exp(-\frac{1}{y}) < y$.

  Hence, $\exp(-\frac{1}{y}) < y < \exp(y)$, so by the intermediate value theorem

$$\exists c \in [-\frac{1}{y}, y] \quad \text{such that} \quad \exp(c) = y.$$

□

Definition. The function $\log : (0, \infty) \to \mathbb{R}$ is the inverse of $\exp : \mathbb{R} \to (0, \infty)$.

Proposition. For $x, y \in (0, \infty)$

- $\log(xy) = \log x + \log y$
- $\log(\frac{x}{y}) = \log x - \log y$

Proof.

- For the first proof

$$\begin{aligned}\exp(\log x + \log y) &= \exp(\log x)\exp(\log y) \\ &= xy \\ &= \exp(\log(xy)).\end{aligned}$$

  so by injectivity $\log xy = \log x + \log y$

- Because $\exp(0) = 1$ this means $\log 1 = 0$.

$$\begin{aligned}\log(\frac{y}{y}) &= \log y + \log(\frac{1}{y}) \\ 0 &= \log y + \log(\frac{1}{y})\end{aligned}$$

so

$$\log(\frac{1}{y}) = -\log y.$$

Hence

$$\begin{aligned}\log(\frac{x}{y}) &= \log x + \log(\frac{1}{y}) \\ &= \log x - \log y.\end{aligned}$$

□

Definition. For $a > 0$ and $x \in \mathbb{R}$

$$a^x = \exp(x \log a).$$

3 Derivatives

Definition. Suppose $S \subseteq \mathbb{R}$. A point $c \in S$ is called an **interior point** of S if $\exists \delta > 0$ such that

$$(c - \delta, c + \delta) \subseteq S.$$

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Suppose $c \in D$ is an interior point of D. We say f is **differentiable** at c if

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \quad \text{exists.}$$

The number $f'(c) = \lim_{h \to 0} \frac{f(c+h)-f(c)}{h}$ is called **the derivative** of f at c. If f is differentiable at every $c \in D$, we say f is **differentiable** on D.

Proposition. Let $D \subseteq \mathbb{R}$ and let $c \in D$ be an interir point of D. If $f : D \to \mathbb{R}$ is differentiable at c, then it is continuous at c.

Proof. Suppose $\lim_{h \to 0} \frac{f(c+h)-f(c)}{h}$ exists. Using a substition of $x = c + h$, this can be rewritten as

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = f'(c)$$

Hence, by the algebra of limits

$$\begin{aligned}\lim_{x\to c}(f(x)-f(c)) &= \lim_{c\to c}\left(\frac{f(x)-f(c)}{x-c}(x-c)\right)\\ &= \lim_{x\to c}\frac{f(x)-f(c)}{x-c}\lim_{x\to c}(x-c)\\ &= f'(c)(0)=0.\end{aligned}$$

so $\lim_{x\to c} f(x) = f(c)$, so f is continuous at c. □

3.1 The Algebra of Derivatives

Theorem. Let $D \subseteq \mathbb{R}$ and let $c \in D$ be an interior point. Suppose that $f, g : D \to \mathbb{R}$ are differentiable at c. Then

- $f+g$ is differentiable at c with
$$(f+g)'(c) = f'(c) + g'(c).$$
- $f-g$ is differentiable at c with
$$(f-g)'(c) = f'(c) - g'(c).$$
- αf is differentiable at c with $(\alpha f)'(c) = \alpha f'(c)$
- (Product rule) fg is differentiable at c with
$$(fg)'(c) = f'(c)g(c) - f(c)g'(c).$$
- (Quotient rule). If $g(c) \neq 0$ then $\frac{f}{g}$ is differentiable at c with
$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c)-f(c)g'(c)}{(g(c))^2}.$$

Proof. The first three points can be proven relatively easily using the algebra of function limits.

- First, it is easy to show that
$$(fg)(x+h)-(fg)(x) = (f(x+h)-f(x))\,g(x+h)+f(x)\,(g(x+h)-g(x))\,.$$
Now, taking the limit of this expression divided by h gives
$$\lim_{h\to 0}\frac{(f(x+h)-f(x))\,g(x+h)}{h}+\frac{f(x)\,(g(x+h)-g(x))}{h}.$$
Since g is differentiable and therefore continuous, $\lim_{h\to 0} g(x+h) = g(x)$, so by the algebra of function limits the product rule is obtained.

- We only need to prove that $\left(\frac{1}{g}\right)'(c) = \frac{-g'(c)}{(g(c))^2}$, and can then apply this with the product rule to prove the quotient rule. We also need to use the continuity of g to prove $g(x+h) \neq 0$ less than some distance δ away. Finally

$$\begin{aligned}\lim_{h\to 0} \frac{\left(\frac{1}{g}\right)(x+h) - \left(\frac{1}{g}\right)(x)}{h} &= \lim_{h\to 0} \frac{g(x) - g(x+h)}{hg(x)g(x+h)} \\ &= \lim_{h\to 0} -\frac{g'(x)}{g(x)g(x+h)}\end{aligned}$$

as required.

□

Corollary. Let $p : \mathbb{R} \to \mathbb{R}$ be a polynomial with

$$P(x) = a_n x^n + ... + a_1 x + a_0.$$

Then P is differentiable on $\mathbb{R}$ and

$$p'(x) = na_n x^{n-1} + ... + a_1.$$

3.2 The Chain Rule

Theorem. Let $D, E \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $g : E \to D$. Let $c \in E$ be an interior point, such that $g(c)$ is an interior point of D. If

- g is differentiable at c
- f is differentiable at $g(c)$
-

then $f \circ g$ is differentiable at c with

$$(f \circ g)'(c) = f'(g(c))g'(c).$$

Proof. Let $h \in \mathbb{R} \setminus \{0\}$ be such that $c + h \in E$. If $g(c+h) \neq g(c)$ then

$$\begin{aligned}\frac{(f \circ g)(c+h) - (f \circ g)(c)}{h} &= \frac{f(g(c+h)) - f(g(c))}{h} \\ &= \frac{f(g(c+h)) - f(g(c))}{g(c+h) - g(c)} \times \frac{g(c+h) - g(c)}{h}.\end{aligned}$$

Define $q : D \to \mathbb{R}$ by

$$q(y) = \begin{cases} \frac{f(y) - f(g(c))}{y - g(c)} & y \in D \setminus \{g(c)\} \\ f'(g(c)) & y = g(c) \end{cases}.$$

Then, if $g(c+h) \neq g(c)$,

$$\frac{(f \circ g)(c+h) - (f \circ g)(c)}{h} = q(g(c+h)) \times \frac{g(c+h) - g(c)}{h}.$$

If $g(c+h) = g(c)$, then it equals 0. Then

$$\begin{aligned}
(f \circ g)'(c) &= \lim_{h \to 0} q(g(c+h)) \lim_{h \to 0} \frac{g(c+h) - g(c)}{h} \\
&= \lim_{h \to 0} q(g(c+h))g'(c) \\
&= \lim_{h \to 0} \frac{f(g(c+h)) - f(g(c))}{f(g(c+h)) - g(c)} g'(c) \\
&= \lim_{h \to 0} \frac{f(g(c+h)) - f(g(c))}{h} g'(c) \\
&= f'(g(c))g'(c).
\end{aligned}$$

□

3.3 The derivative of inverse functions

Theorem. Let $D, E \subseteq \mathbb{R}$ and let $f : D \to E$ bijective function. Suppose $c \in D$ and $f(c) \in E$ are interior points. If f is differentiable at c with $f'(c) \neq 0$ then f^{-1} is differentiable at $f(c)$ with

$$(f^{-1})'(f(c)) = \frac{1}{f'(c)}.$$